VITyarthi | Open Source Software

The Open Source Audit

A Capstone Project for OSS NGMC Course

Course: Open Source Software |

Unit Coverage: 1 – 5 Max

Marks: 100

Chosen Software: GIT

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Slot	**Date of Submission**
B22	**31-03-2026**

1. Introduction

The majority of software applications which people use now originate from developer groups who collaborate instead of being developed by individual companies. Open-source software exists to fulfill this purpose. The platform enables users to distribute their creations. They can enhance them and access all features without any cost.

I selected Git as my open-source software choice for this project. Git functions as a version control system which enables developers to monitor code modifications while working together with other developers. The system has become a standard tool that software development teams use for their projects.

The original purpose of Git was to address specific challenges which Linux developers encountered during their programming work. The software gained widespread acceptance because users found its performance capabilities to be fast and its operational functions to be adaptable. The software has become an open source project which enables developers from every country to participate in its development work.

The report examines Git through various lenses which include its historical background and licensing details together with its operational capabilities on Linux systems and its function within the open-source software community. The comparison will be made between Git and proprietary software systems.

I created five shell scripts to demonstrate basic Linux commands and automation which I included with my report. The scripts provided me with practical knowledge about how open-source tools function.

2. Choosing Your Software

Your choice	Write your chosen software here before submission: Git

Part A — Origin and Philosophy (Units 1 & 2)

A1. Origin Story

Git was created in 2005 by Linus Torvalds, who also developed the Linux kernel. The Linux project used BitKeeper, which was a proprietary tool, to manage their code changes before they switched to Git.

BitKeeper became inaccessible to the Linux community because of licensing problems which prevented its use. The situation created major difficulties because the Linux kernel development team required an effective system to control their work.

The existing version control systems at that time showed two main problems because they either operated too slowly or failed to support distributed development. Linus Torvalds created his own version control system because he wanted a better solution.

He created the Git system to achieve three goals which included fast operation and efficient performance and capability to manage projects of any size. The Git system contains a crucial function which enables all developers to access complete project files through its distributed system.

Linus released Git as open source under the GPL license because he wanted to make it accessible instead of keeping it as a private project. The release provided worldwide developers the ability to use the system which they could then make improvements to. This development led to Git becoming the most widely used version control system.

A2. License: what it actually says

Git is licensed under the GNU General Public License version 2 (GPL v2). This license is designed to give users complete freedom over the software.

There are four main freedoms in open-source software:

- The freedom to use the software for any purpose
- The freedom to study how it works and modify it
- The freedom to share copies with others
- The freedom to distribute modified versions

Git follows all these freedoms. This means anyone can use Git, modify it, and even share their modified versions.

One important feature of the GPL license requires anyone who modifies and distributes software to provide its complete source code. This provision guarantees perpetual accessibility of the software to the public.

This is different from licenses like MIT, where companies can modify the software and keep their changes private. In my opinion, GPL is better for projects like Git because it protects the open-source nature.

It is not only 'open-source'- that puts in the position of the user in charge-but 'control'.

A3. The ethics of open source

Open-source software raises an interesting question: should all software be open source?

Open source software enables people to see all system elements while allowing knowledge sharing between users. The security of the system increases because users have access to its codebase for examination. The system provides students and developers with opportunities to study actual projects. The financial requirements for open source software development exceed the capabilities of some companies. The companies need to make money because their projects require extensive funding to keep operations going.

Google and Microsoft use the Linux operating system as an open-source software solution. Some individuals consider this practice to be unjust yet the companies still give back to their community.

The idea of "standing on the shoulders of giants" is very important here. Developers today do not start from scratch. They build on existing tools like Git, Linux, and Python.

In my opinion, open source supports innovation rather than limiting it. It saves time and allows developers to focus on new ideas instead of reinventing things.

Part B — Linux Footprint (Unit 2)

Git operates very efficiently in Linux environments and is commonly available by default in many distributions.

If it isn't already installed, you can easily install it using a package manager command such as:

sudo apt install git

After installation, the Git executable is typically found at /usr/bin/git, while its configuration settings are stored in the user's home directory, usually in a file like ~/.gitconfig.

Git works at the user level, meaning it doesn't require root privileges for everyday tasks. This makes it both secure and convenient to use.

Unlike some applications, Git does not run continuously in the background-it is only active when you execute commands in the terminal.

Updates and new features for Git are distributed through package managers, with ongoing improvements contributed by the open-source community.

Part C — The FOSS Ecosystem (Units 3 & 4)

Git plays a very important role in the open-source ecosystem.

The platforms GitHub and GitLab and Bitbucket which support worldwide developer project collaboration use this tool. Git requires basic system tools for its operation because it functions as an independent system while providing support to additional software and services.

A vast community of developers actively participates in Git development by fixing bugs and creating new features and enhancing existing performance capabilities. Git operates as a web-based tool which most code hosting platforms use to manage their code repositories.

Part D — Open Source vs Proprietary (Critical Analysis)

Dimension	Open Source	Proprietary Alternative
Cost	Git is completely free to use, download, and modify.	Proprietary tools often require paid licenses, especially for enterprise usage.
Security (who can audit the code?)	The source code is publicly available, so anyone can review it and find vulnerabilities.	The source code is closed, so only the company can check and fix security issues.
Support and reliability	Supported by a large global community with extensive documentation, forums, and frequent updates.	Provides official customer support, but depends on subscription and company response time.
Freedom to modify	Users can freely modify Git and even create their own versions according to their needs.	Users cannot modify the software and must follow company-imposed restrictions.
Community vs corporate control	Developed and maintained by a global open-source community where contributions are shared.	Controlled by a company, where decisions are driven by business goals.
Your overall verdict	Git is flexible, free, and widely used, making it ideal for developers and collaborative projects.	Proprietary tools may offer structured support but lack flexibility and freedom.

Conclusion

From my perspective Git serves as the superior option for most developers because it provides a free solution which supports multiple development needs and has extensive community backing. The system enables users to manage all aspects of their codebase while fostering teamwork between developers. Organizations that use proprietary software solutions gain access to specialized customer service but they must pay high expenses while dealing with multiple limitations on product use and system adaptability.

4. Shell Script Tasks

Script 1 — System Identity Report (Unit 1 & 2)

```
#!/bin/bash
# Script 1: System Info Summary
# Author: Roshan Gayakwad

NAME="Roshan Gayakwad"
TOOL_USED="Git"

KERNEL_VER=$(uname -r)
CURRENT_USER=$(whoami)
CURRENT_TIME=$(date "+%d-%m-%Y %H:%M:%S")

# Uptime check
if command -v uptime >/dev/null 2>&1; then
    SYS_UPTIME=$(uptime -p)
else
    SYS_UPTIME="Unavailable"
fi

# OS Detection
if [ -f /etc/os-release ]; then
    OS_NAME=$(grep PRETTY_NAME /etc/os-release | cut -d= -f2 | tr -d '")
else
    OS_NAME="Unknown / Git Bash"
fi

echo "--------------------------------"
echo "   SYSTEM REPORT DASHBOARD"
echo "--------------------------------"
echo "Student Name : $NAME"
echo "Tool Used    : $TOOL_USED"
echo "User         : $CURRENT_USER"
echo "Kernel Ver   : $KERNEL_VER"
echo "Operating OS : $OS_NAME"
echo "System Uptime: $SYS_UPTIME"
echo "Timestamp    : $CURRENT_TIME"
echo "License Info : GPL (Linux)"
echo "--------------------------------"
```

```
ROSHAN@Roshans MINGW64 /f/download/oss-audit-24bce10410-main/oss-audit-24bce1041
0-main/Scripts
$ ./script1.sh
--------------------------------
   SYSTEM REPORT DASHBOARD
--------------------------------
Student Name : Roshan Gayakwad
Tool Used    : Git
User         : ROSHAN
Kernel Ver   : 3.6.6-1cdd4371.x86_64
Operating OS : Unknown / Git Bash
System Uptime: Unavailable
Timestamp    : 31-03-2026 01:05:10
License Info : GPL (Linux)
```

Script 2 — FOSS Package Inspector (Unit 2)

```
#!/bin/bash
# Script 2: Software Checker
# Author: Roshan Gayakwad

PKG="git"

echo "Verifying installation of $PKG"
echo "================================"

if command -v $PKG >/dev/null 2>&1; then
    echo "$PKG is available on this system."
    echo "Installed Version: $($PKG --version)"
else
    echo "$PKG is missing."
fi

# Info block
case $PKG in
    git) echo "Git is a distributed version control system." ;;
    vlc) echo "VLC is a media player." ;;
    firefox) echo "Firefox is a web browser." ;;
    *) echo "No additional info found." ;;
esac
```

```
ROSHAN@Roshans MINGW64 /f/download/oss-audit-24bce10410-main/oss-audit-24bce1041
0-main/Scripts
$ ./script2.sh
Verifying installation of git
================================
git is available on this system.
Installed Version: git version 2.53.0.windows.1
Git is a distributed version control system.
```

Script 3 — Disk and Permission Auditor (Unit 2)

```
#!/bin/bash
# Script 3: Directory & Permission Checker
# Author: Roshan Gayakwad

FOLDERS=("/etc" "/usr/bin" "/tmp" "$HOME")

echo "==== Directory Inspection ====="
echo ""

for F in "${FOLDERS[@]}"; do
    if [ -d "$F" ]; then
        DETAILS=$(ls -ld "$F" | awk '{print $1, $3}')
        SIZE_INFO=$(du -sh "$F" 2>/dev/null | cut -f1)
        echo "Path: $F"
        echo "Permissions & Owner: $DETAILS"
        echo "Size: $SIZE_INFO"
        echo "----------------------------"
    else
        echo "$F does not exist"
    fi
done

# Git config check
if [ -f "$HOME/.gitconfig" ]; then
    echo "Git configuration detected:"
    ls -l "$HOME/.gitconfig"
else
    echo "Git config file not found."
fi
```

```
ROSHAN@Roshans MINGW64 /f/download/oss-audit-24bce10410-main/oss-audit-24bce10410-main/Scripts
$ ./script3.sh
==== Directory Inspection ====

Path: /etc
Permissions & Owner: drwxr-xr-x ROSHAN
Size: 1.6M
----------------------------
Path: /usr/bin
Permissions & Owner: drwxr-xr-x ROSHAN
Size: 91M
----------------------------
Path: /tmp
Permissions & Owner: drwxr-xr-x ROSHAN
Size: 1.5G
----------------------------
```

Script 4 — Log File Analyzer (Units 2 & 5)

```
#!/bin/bash
# Script 4: Log Keyword Scanner
# Author: Roshan Gayakwad

FILE_PATH=$1
SEARCH_TERM=${2:-"error"}
MATCH_COUNT=0

if [ ! -f "$FILE_PATH" ]; then
    echo "Invalid file path!"
    exit 1
fi

while IFS= read -r TEXT; do
    if echo "$TEXT" | grep -iq "$SEARCH_TERM"; then
        ((MATCH_COUNT++))
    fi
done < "$FILE_PATH"

echo "The term '$SEARCH_TERM' occurred $MATCH_COUNT times."

echo ""
echo "Recent matches (last 5):"
grep -i "$SEARCH_TERM" "$FILE_PATH" | tail -n 5
```

```
ROSHAN@Roshans MINGW64 /f/download/oss-audit-24bce10410-main/oss-audit-24bce10410-main/Scripts
$ ./script4.sh script1.sh echo
The term 'echo' occurred 11 times.

Recent matches (last 5):
echo "Operating OS : $OS_NAME"
echo "System Uptime: $SYS_UPTIME"
echo "Timestamp    : $CURRENT_TIME"
echo "License Info : GPL (Linux)"
echo "--------------------------------"
```

Script 5 — The Open Source Manifesto Generator (Unit 5)

```
#!/bin/bash
# Script 5: Personal Open Source Note
# Author: Roshan Gayakwad

echo "Fill in the details below:"

read -p "Your favorite tool: " TOOL_NAME
read -p "Meaning of freedom: " MEANING
read -p "What you aim to create: " GOAL

TODAY=$(date "+%d %B %Y")
OUTPUT_FILE="oss_note_$(whoami).txt"

{
echo "===== Open Source Reflection ======"
echo "Date: $TODAY"
echo ""
echo "For me, open source stands for $MEANING."
echo "Tools like $TOOL_NAME empower developers."
echo "I plan to build $GOAL and share it with the community."
} > "$OUTPUT_FILE"

echo "File saved as: $OUTPUT_FILE"
echo ""
cat "$OUTPUT_FILE"
```

```
ROSHAN@Roshans MINGW64 /f/download/oss-audit-24bce10410-main/oss-audit-24bce10410-main/Scripts
$ ./script5.sh
Fill in the details below:
Your favorite tool: Unity
Meaning of freedom: Free
What you aim to create: Games
File saved as: oss_note_ROSHAN.txt

===== Open Source Reflection =====
Date: 31 March 2026

For me, open source stands for Free.
Tools like Unity empower developers.
I plan to build Games and share it with the community.
```